

Logistic Regression

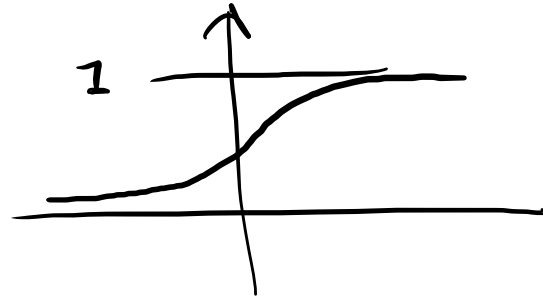
Binary classification

{ setosa
virginica } ← iris

{ True or
False } ← Cancer?

Sigmoid:

$$\frac{1}{1 + e^{-x}}$$



multiclass classification:

→ softmax: $\frac{e^{x_i}}{\sum e^{x_i}}$

softmax for Binary class.

$$\text{softmax}(x_1) = \frac{e^{x_1}}{e^{x_1} + e^{x_2}}$$

$$\text{softmax}(x_2) = \frac{e^{x_2}}{e^{x_1} + e^{x_2}}$$

$$\text{softmax}(x_1) + \text{softmax}(x_2) = 1$$

$$\frac{e^{x_1}}{e^{x_1} + e^{x_2}} + \frac{e^{x_2}}{e^{x_1} + e^{x_2}} = \frac{e^{x_1} + e^{x_2}}{e^{x_1} + e^{x_2}} = 1$$

$$\frac{e^{x_1}}{e^{x_1} + e^{x_2}} = \frac{\quad}{\quad}$$

$$\frac{e^{x_2}}{\quad} \rightarrow \frac{e^{x_1}}{\quad}$$

$$\frac{1}{1 + e^{-x}}$$

$$\frac{e^{x_1}}{e^{x_1} + (\alpha e^{x_1})} \rightarrow$$

$$\frac{1}{1 + e^{-x_1}}$$

	a	b	c	
	<u>70</u>	40	40	→ <u>150</u>
acc	70	40	40	→ 150

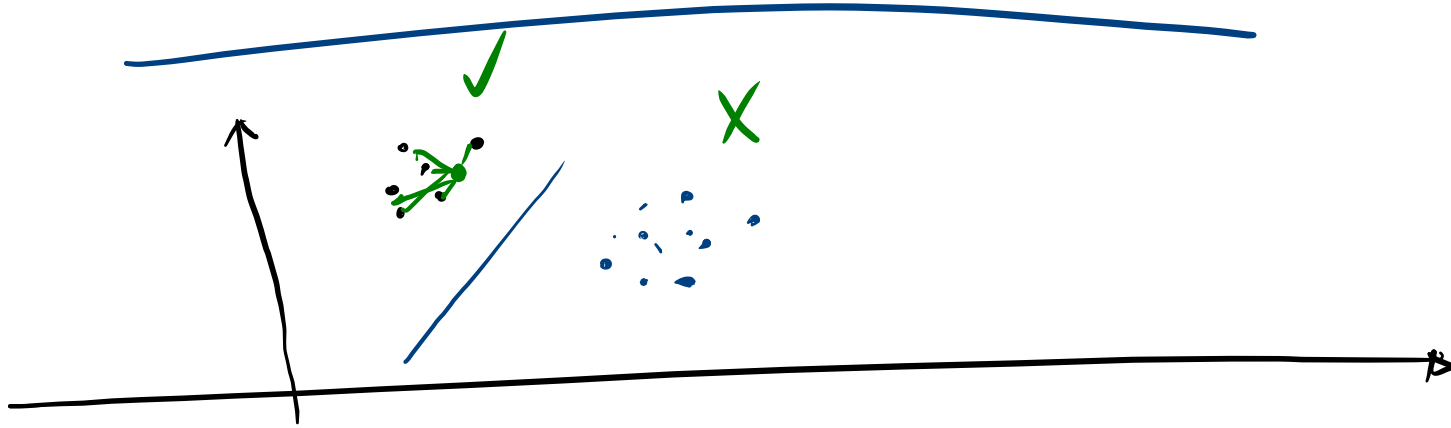
0.90	0.95	0.80	0.40
			$\frac{40}{150}$

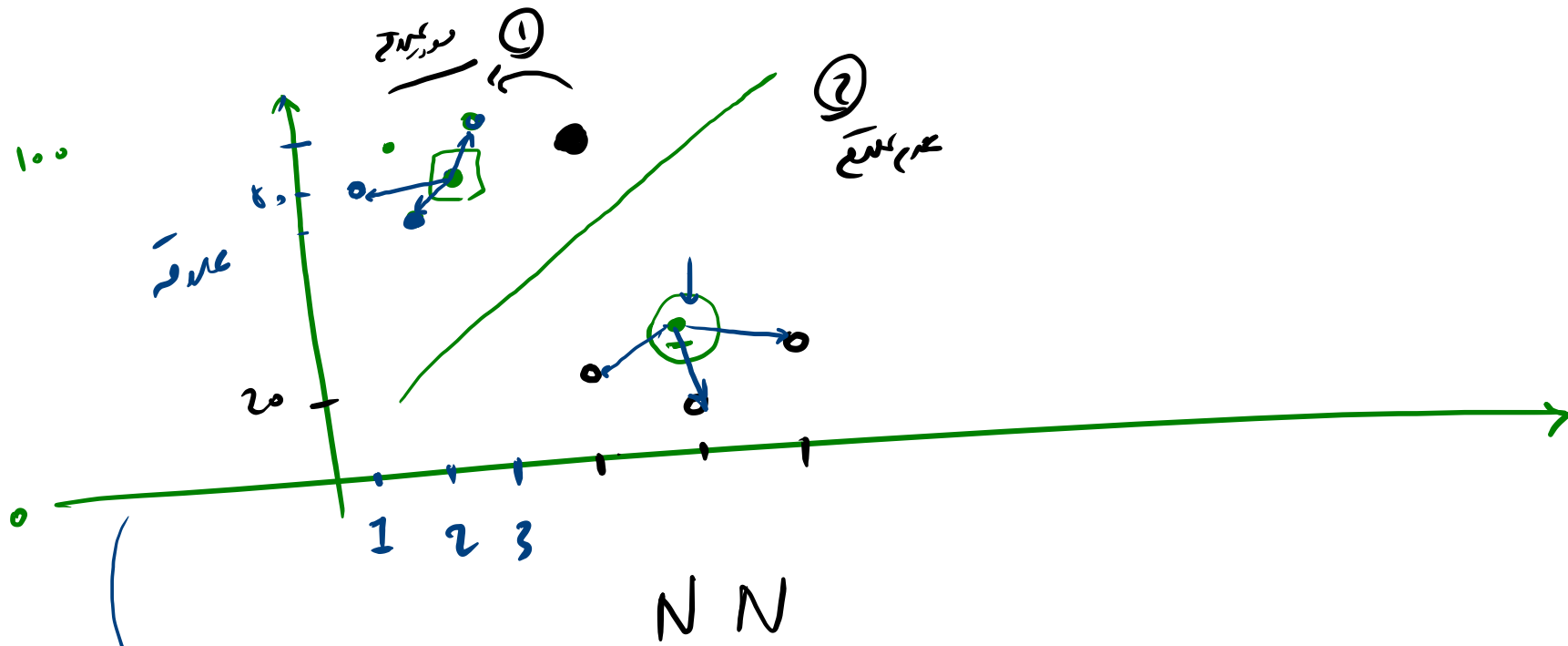
weight:

$$w_1 acc_1 + w_2 acc_2 + w_3 acc_3$$

$$w_1 + w_2 + w_3 = 1$$

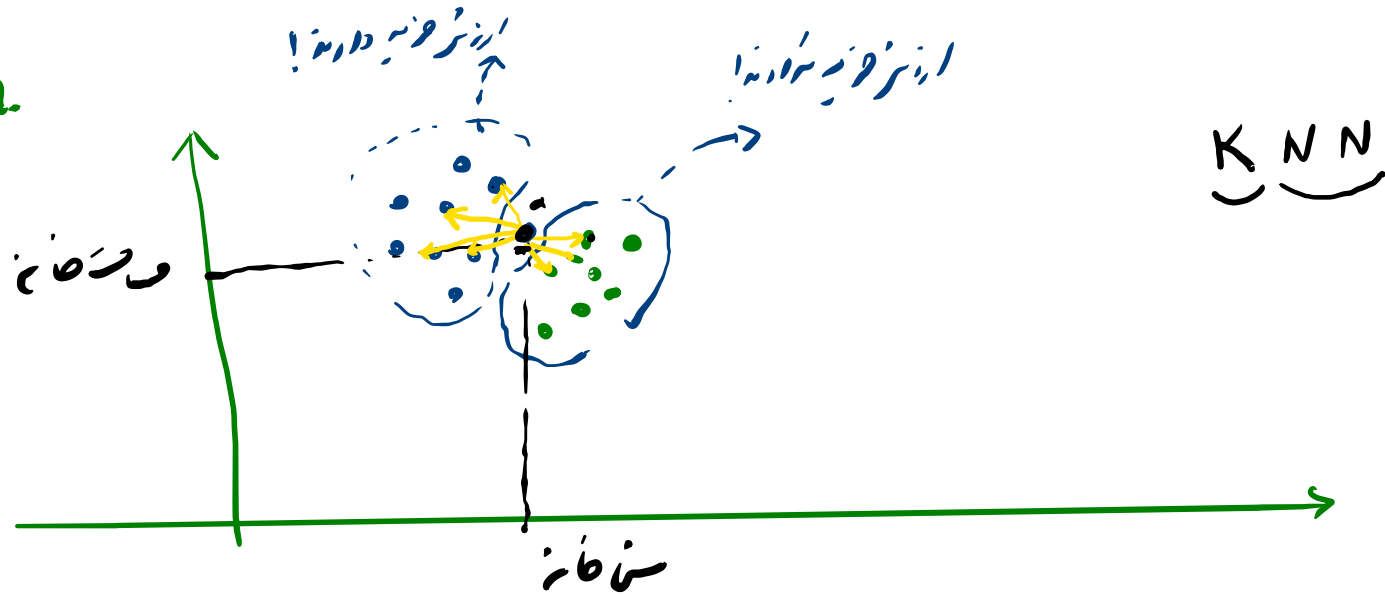
کچھ ازاد سُر ← ہر سِدیں ازاد سُر لاکر سِلتے ہی منہ لے جے بہ فرما ہے!





nearest neighbors

- : class 1
- : class 2

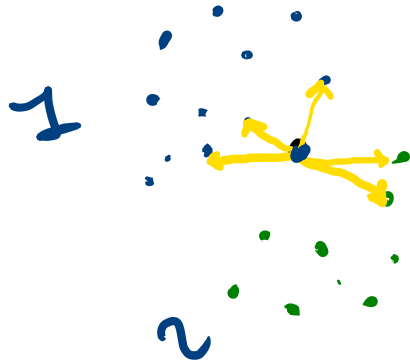


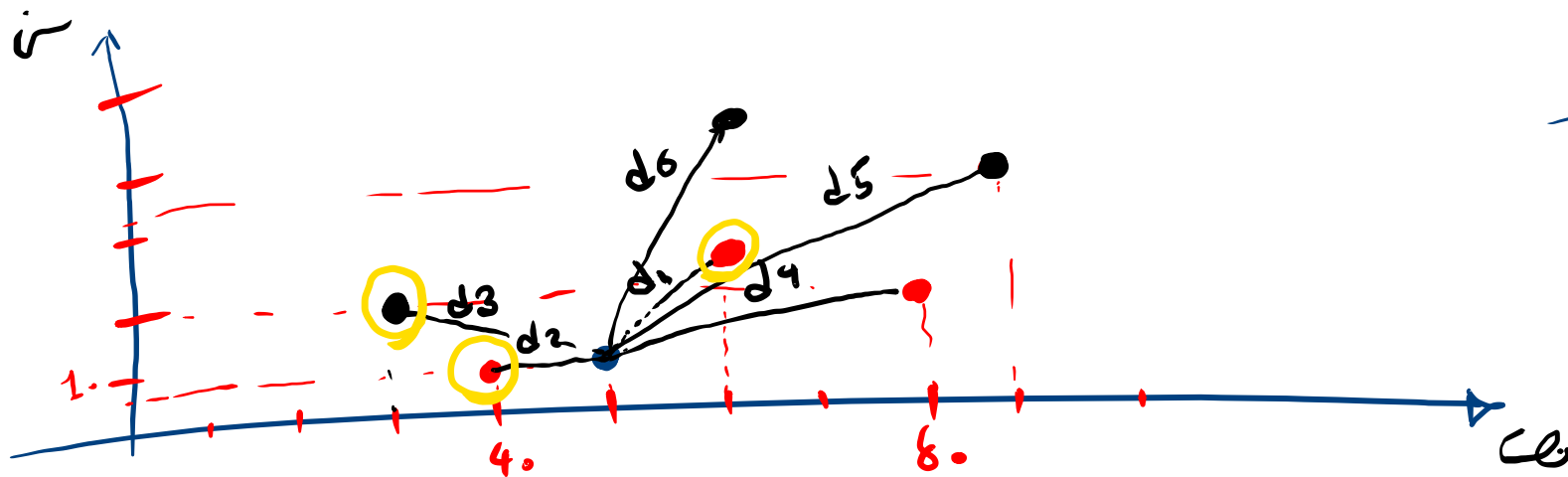
$$\sqrt{(x_j - x_i)^2 + (y_j - y_i)^2}$$

5NN



3 : class 1 → class 1
2 : class 2





KNN: $k=3$: ساختار

$NN: [d_1, d_2, d_3]$
True True False

2: True ✓
 1: False

io c	io i	نتیجه
7.0	2.0	True
1.0	1.0	True
4.0	1.0	True
9.0	2.0	False
4.0	2.0	False
7.0	1.0	False
unseen	5.0	?
	1.0	?

True: ●

False: ○

اگر α نزدیک KNN ← ما وزن می دهیم، بعد از آن استفاده می کنیم.
k nearest neighbor.

Train
می دهیم!

the end